

Statistical Methods Quiz #3 (50 minutes)

'26. 05. 22. (Friday) Cadet number(교번): _____ Name: _____ Score: _____

<Instructions>

1. It is open-book, closed-web test.
2. You are not allowed to discuss on the problems during the exam.
3. Import *school.csv* data first. (Use all the string variables as factors.)

sex: Gender. F = female, M = male

age: Age. Numeric values ranging from 15 to 22

Pstatus: Parents' cohabitation status. T = living together, A = apart

Medu: Mother's education level. 0 = none, 1 = primary education, 2 = grades 5-9, 3 = secondary education, 4 = higher education

Fedu: Father's education level. 0 = none, 1 = primary education, 2 = grades 5-9, 3 = secondary education, 4 = higher education

guardian: Primary guardian. Mother, father, or other

traveltime: Travel time to and from school. 1 = less than 15 min, 2 = 15-30 min, 3 = 30 min-1 hour, 4 = more than 1 hour

studytime: Study time. 1 = less than 2 hours, 2 = 2-5 hours, 3 = 5-10 hours, 4 = more than 10 hours

failures: Number of past class failures. 1, 2, 3, or 4+

freetime: Free time. Ordinal value from 1 (very low) to 5 (very high)

famrel: Family relationship quality. Ordinal value from 1 (very bad) to 5 (excellent)

1. Import *school.csv* dataset. (Activate "Strings as factors" option) Use the **str** function to examine the structure of the *school* data. How many observations and how many variables does the dataset contain? [15 points]

<R-code>

<Output>

The number of observations: _____

The number of variables: _____

2. If a student's "absences" variable is larger than 9, the student is classified as being "lazy". Define a factor variable "lazy" in *school* dataset using **ifelse** and **factor** function. (with levels "Yes" and "No") [15 points]

<R-code>

3. Using the *school* data, fit a **classification tree** with **lazy** as the response variable and all other variables **except absences** as explanatory variables. Also, visualize the fitted tree. What is the variable used in the first split? [15 points]

<R-code>

<Output (plot)>

The variable used in the first split: _____

4. Using the *school* data, obtain the predicted response variable. Then, create a **confusion matrix** and compute the **misclassification error rate**. [20 points]

<R-code>

<Output>

Misclassification error rate: _____

5. Using the school data, fit a **random forest** model with **lazy** as the response variable and all other variables **except absences** as explanatory variables. (Set the number of variables randomly selected at each split to **3**.) Using the data, obtain the predicted response variable. Then, create a **confusion matrix** and compute the **misclassification error rate**. Based on misclassification error rate, which model performs better for the training data: the **single decision tree** or the **random forest**? (Select one.) [20 points]

<R-code>

<Output>

Misclassification error rate: _____

The (single decision tree / random forest) model has better performance.

6. For the random forest model, visualize the **variable importance**. Which variable is the most important? [15 points]

<R-code>

<Output (plot)>

The most important variable: _____